

# AI Agent Knowledge Base for a Financial Company

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## The Principles

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Every decision in this document traces back to one of these. They are listed here so a reader can see the whole argument in a minute, and so a reviewer can check any section against the rule it claims to follow. **The first three are the ones everything else hangs from**; the rest follow from them.

1. **Simplicity above all.** The fewest moving parts that still deliver the full functionality. In particular, **no distributed architecture**: one server rather than a cluster, a mounted Unix filesystem rather than an object store, a database table rather than a message broker.

2. **Modularity.** Every capability behind a contract (API, plugin, port), with loose coupling, encapsulation, and bounded contexts, so any part can be changed, replaced, or tested on its own. The same discipline reaches into the code: subdirectories per module, files under 800 lines, functions under 50, docs at every level, a `README.md` per directory.
3. **AI as a component, not a feature.** Agents run through the whole system and through the process that designed it, each with a knowledge base it must cite, a memory scoped to one identity, and tools reachable only through the audited APIs. The agent is available to **every user and every maintainer**: talk to it to build mini-tools and workflows, schedule and run jobs, run analyses, and draft reports and documents.
4. **Self-healing.** Timeouts and retries, snapshots and restore, degraded indexes rebuilt, derived artifacts regenerated. **A restart is a non-event**, because no durable state lives in process memory and every job resumes from the queue. Humans get paged as needed, never for the routine.
5. **One system of record.** PostgreSQL holds all transactional truth. Everything else is a copy, a cache, or a projection, and is labeled as such.
6. **Parse once.** The expensive work happens a single time; everything downstream is disposable and cheap to regenerate.
7. **Search three ways, more than once.** Semantic vectors for meaning, BM25 for the exact string, the document graph and each long document's own table-of-contents tree for structure, with time as a dimension across all three, and a bounded multi-step loop so the agent can go back for what the first pass missed. **Retrieve small, relevant, and authorized**, rather than pouring the corpus into a context window.
8. **Security and provenance are load-bearing.** ACLs filter in SQL before ranking; every claim carries a verified citation; every answer leaves an immutable audit trail; "I don't know" is a graded skill.
9. **Verify continuously.** Unit, module, integration, and AI conformance tests gate every merge. A golden-suite evaluation gate, reconciliation counts, and hallucination sampling gate every day.
10. **No failure is orphaned.** Every escalation becomes a monkey: a task with an owner, a status, and a history. People manage and hand off their own, escalate what's above their pay grade, and managers see the whole troop.
11. **Guardrails on generated work.** Anything an AI generates reaches data only through the documented, audited APIs, so entitlements and audit apply automatically, and it passes the same CI, size, and conformance gates as human-written code.
12. **Nothing arrives unvetted.** Dependencies are pinned, aged 30 days, and verified. "Latest" is not a version, it's a gamble.

The rest of this document is the story of how each principle earned its place, told as the problems that forced it. ["The Shape of the Thing"](#) at the end distills what the principles produced.

## The Problem Nobody Wants to Own

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Somewhere in the firm's file shares, SharePoint sites, mail archives, and document management systems sit roughly **three million documents**: credit agreements, amendments, side letters, board decks, risk memos, spreadsheets, and thirty years of scanned paper. Somewhere in there is the answer to almost any question a lawyer, credit officer, or risk analyst might ever ask.

Nobody can find it.

Keyword search cheerfully returns four hundred hits, which is a polite way of returning zero. The one person who knew which document actually mattered retired in the spring. And the version someone finally digs up? Superseded two amendments ago: a fact that will surface at the worst possible moment, because that is when such facts surface.

So the ask sounds simple: *build an AI assistant that answers questions from our documents*. Lovely. Except this is a regulated financial firm, so the real ask comes with fine print:

- Answer from the documents, and **prove it**: citations down to the page, tied to the exact version used. "Trust me" is not a citation.
- Show a user **only what they're entitled to see**. Not one chunk more. Ethical walls between deal teams are law, not office etiquette.
- Keep everything (documents, indexes, models, logs) **inside the firm's walls**. Nothing takes a field trip to someone else's cloud.
- Keep a **tamper-evident record** of every question and answer, because a regulator may ask about it years from now, and "we're not sure what it said" is a career-limiting answer.
- And when the system doesn't know, it must **say so**, not improvise something plausible in a confident voice.

That last point deserves a moment of silence. A chatbot that hallucinates a covenant threshold isn't a productivity tool; it's a liability engine with a friendly interface. Everything in this design flows from taking that seriously.

One deliberate act of modesty: version 1 is **read-only**. It answers questions. It executes no trades, sends no emails, makes no decisions. Walk before you run, especially in a building full of compliance officers.

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## Why Not Just Paste It All Into the Model?

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Two shortcuts get proposed at the start of every project like this one, and both were respectable positions not long ago.

**The first is one-shot vector RAG.** Embed the corpus, embed the question, pull the top twenty chunks, paste them into a prompt, ship it. Through 2023 and 2024 that was the entire playbook, and it spread everywhere because it demos beautifully. It fails in a specific and predictable way: one shot means one guess. Ask "*which amendments changed the leverage covenant, and what did it say before?*" and no single search can answer it, because the second search depends on what the first one found. A lone retrieval pass cannot ask a follow-up question.

By 2026 the field has moved to **agentic, multi-step retrieval**. The system searches, reads what came back, notices what's missing, and searches again, which is what a competent analyst does with a filing cabinet and always has.

**The second shortcut is newer and far more seductive: skip retrieval altogether.** Context windows are enormous now. Why not hand the model the whole credit agreement, or the whole deal folder, and let it sort out the rest?

Start with arithmetic. This corpus runs past **ten billion tokens**, and the largest windows on the market measure in the millions. The whole corpus was never going in, so the only real question is who does the selecting and how well. Even the modest version, stuffing one deal folder into every prompt, means paying for a hundred thousand tokens to answer a question that four thousand tokens would have answered, on every query, all day, forever.

Accuracy is the better argument anyway. **A big window is not a free lunch.** Evidence sitting in the middle of a very long context gets skimmed rather than read, and the effect gets worse as the window grows. Forty well-chosen chunks beat four hundred mediocre ones, and they beat them on the exact questions that matter, the ones where the answer hinges on a single clause.

Then comes the argument that ends the discussion in a regulated firm: **a context window has no access control.** Whatever goes in can come out. "We put everything in the prompt and asked the model to be careful about ethical walls" is not a control, it's a wish with a budget. Pasted text has no version, no page number, and no provenance either, so the citations come back as plausible-looking strings rather than as links to the exact paragraph a regulator can open.

**So the design is a knowledge runtime, not a vector database with a chat box bolted to it.** Keyword search for the exact string, vectors for meaning, the document graph for structure, and the derivative store for the file itself, all behind one retrieval API that an agent may call more than once. Every pass filters by entitlement before it ranks. Every claim comes back with a citation to a version and a page. Contradictions and superseded text get surfaced instead of quietly averaged, and every answer carries a confidence signal the reader can see.

Which gives the rule the rest of this document keeps returning to: **retrieve small, relevant, and authorized.** Not everything the model could physically hold. Just what the question needs, from what this particular person is allowed to read.

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## The Temptation of Shiny Infrastructure

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The next challenge isn't technical either. It's saying no.

The standard architecture for a system like this, the one in every vendor deck, involves a vector database, a search cluster, a message broker, a workflow engine, and a graph database. Collect all five! Each one is a distributed system to operate, patch, secure, and keep synchronized with the others. Every pair of systems is a place where data can quietly disagree. Every extra cluster is a pager that will eventually go off at 2 a.m., and it will not be sorry.

This is how projects die, by the way. Rarely from missing features; almost always from drowning in their own plumbing, sometimes before the first user asks the first question.

So this design adopts **simplicity as its guiding principle**: the simplest, most elegant architecture that still delivers the full functionality: the minimum number of moving parts, each one chosen on purpose. Simplicity is what makes the project *do-able*, and flexible, maintainable, and affordable. We can prototype fast, put something real in front of users in weeks, and run the whole thing without a big team, a fleet of clusters, or exotic hardware.

And here's the delicious part: the numbers say we can get away with it. Our corpus, big as it feels, produces about **20 to 25 million searchable chunks**. That is not "big data." For a well-fed PostgreSQL instance, that's a Tuesday.

**So the solution is a bet on gloriously boring technology**: one PostgreSQL server holds everything transactional: the document registry, the entitlements, the search indexes (vector *and* keyword), the job queue, and the audit log. The team already knows how to run PostgreSQL. Access control becomes a SQL join instead of a distributed-systems research project. There is exactly one source of truth, and everyone knows its address.

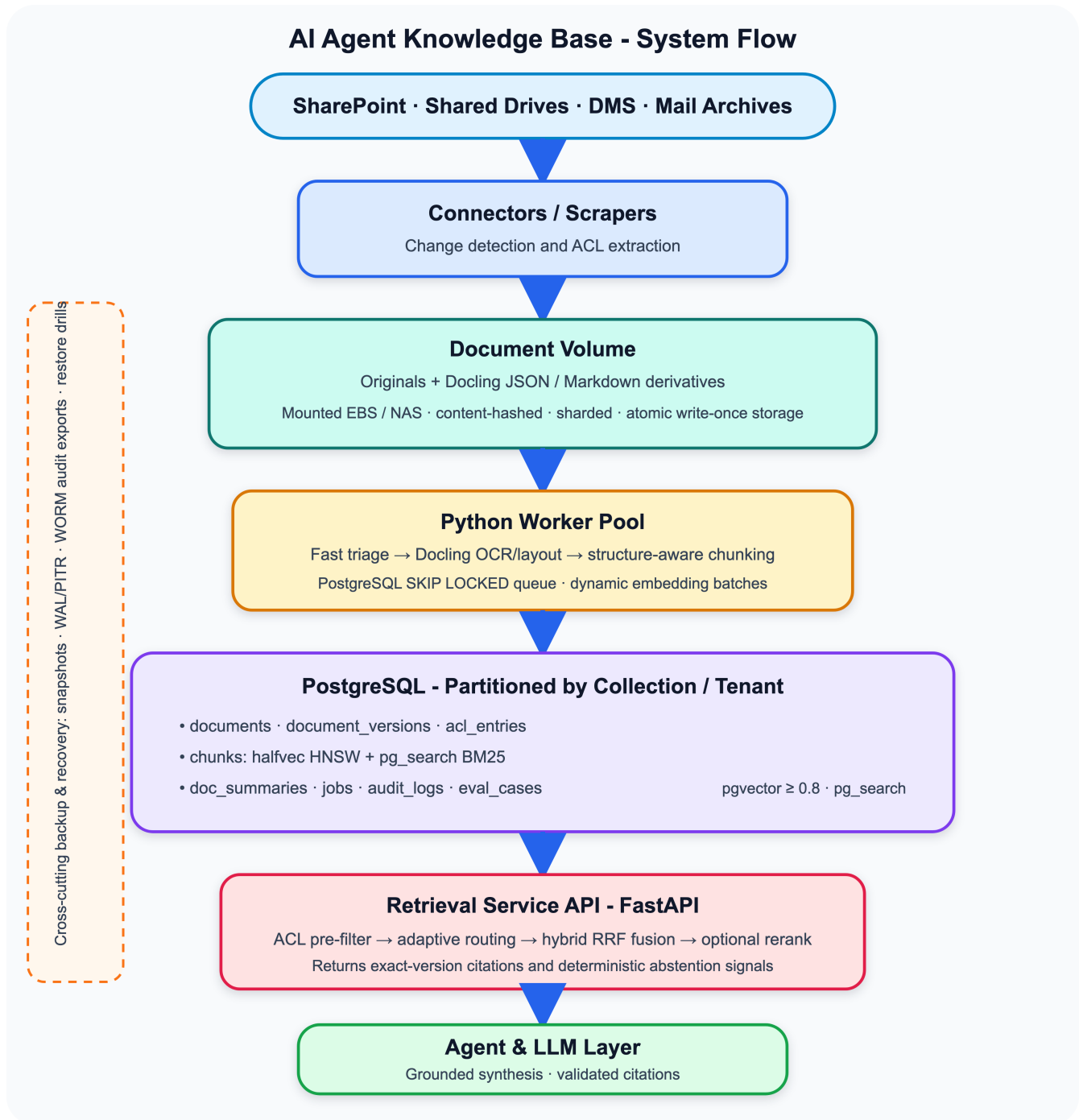
Which brings out the sharp edge of the simplicity principle: **avoid distributed architectures**. One server, not a cluster. A standby exists for failover and it does nothing but wait, because that is high availability, not distribution. Documents live on a mounted Unix filesystem, not in an object store, so a parser opens a file instead of negotiating with an API over the network. Nothing here is sharded, gossiped, quorum-voted, or eventually consistent. Every one of those words is a research project wearing a product name, and each buys a class of failure that only shows up in production, at scale, on a holiday weekend. A single server handles this corpus with room to spare, and the machine that isn't in the diagram never pages anybody.

Because bets should be falsifiable, the design writes down **scaling gates**: measurable thresholds (chunk count above 50 million, p95 latency above 2 seconds despite tuning, index churn degrading reads) at which we'd graduate to specialized infrastructure. Until a gate trips, we don't. When one does, the migration is a background re-index, not a crisis, for reasons that will become clear shortly.

**The same restraint picks the language: Python, not Rust**. The question comes up in every design review, so let's answer it here. A Rust rewrite makes the code faster. But look at where this system actually spends its time: search runs inside PostgreSQL (written in C), embeddings and reranking run on GPUs (CUDA kernels), PDF parsing runs in PyMuPDF (C bindings) and Docling's native internals. The Python code is a thin orchestration layer: it moves job rows, calls libraries, and serializes API responses. Rewriting the glue in Rust makes the glue faster, and the glue was never the bottleneck.

Meanwhile the costs would be real. The document-understanding and ML ecosystem is Python-first. Docling *is* Python, and the best parsing, OCR, and model tooling all live there. The team knows Python; prototypes ship in days, and the pool of people who can maintain the system stays large. Choosing Rust here would mean either wrapping the same Python/C libraries in FFI (all the complexity, none of the benefit) or abandoning the best tools for the job, trading the crown jewels for type safety in the plumbing.

And modularity keeps even this decision reversible: if some module ever proves CPU-bound under *measurement* (chunking or deduplication at a much larger scale, say), that one module can be reimplemented in Rust behind the same contract, invisible to everything else. No crisis, no rewrite; a swap. Though the honest first answer to a slow worker is usually even more boring: add another worker node. Hardware is cheaper than a second language in the team's stack.



## Don't Let the Dumplings Fuse

Simplicity has an evil twin, and it shows up wearing simplicity's clothes.

Think of a good system as a plate of dumplings. There's nothing wrong with a dumpling. A dumpling is a small masterpiece of engineering. Each one has a wrapper that keeps its filling to itself (*encapsulation*), each one holds its own distinct filling (*separation of concerns*, a *bounded context* in a doughy jacket), each can be

cooked and tasted on its own (*developed and tested separately*), and they sit loosely together on the plate, touching but never sticking (*loose coupling*). That is exactly the architecture we want.

The failure mode is what happens when you boil them too long and walk away. It goes like this. A small team builds a lean system: one database, a handful of components, nothing wasted. Then, under deadline, the agent code starts querying the chunks table directly (why bother with an API for an internal tool?). The parser learns which embedding model is in use (it was convenient). Every shortcut is individually reasonable. Two years later the dumplings have **fused into a single messy clump**. The diagram still shows few components, but you can no longer change *any* of them without understanding *all* of them, and nothing can be developed or tested on its own anymore. The system is small, and it is stuck.

That failure mode matters doubly here, because this design makes bets (one database for everything, self-hosted models, a specific parser) and bets must stay reversible. The scaling gates promise "when a threshold trips, we migrate calmly." That promise is only honest if the thing behind the gate can be swapped without a rewrite.

**So the second guiding principle, alongside simplicity, is modularity: every capability owns its domain and is used only through its contract.** (The third one is coming in the next section, and it has opinions about memory.)

- **Retrieval is a service, not a schema.** Agents and internal systems talk to the FastAPI retrieval API; not one of them knows a table name. The day the chunk index moves to a dedicated search cluster, consumers won't even notice.
- **Connectors are plugins.** Each source system speaks its own dialect on one side and emits the same normalized events (file, metadata, ACL) into the same job queue on the other. Adding a source means writing an adapter, not performing surgery on the pipeline.
- **Parsers hide behind the derivative contract.** Docling and PyMuPDF both produce the same structured JSON + Markdown shape. A better parser next year changes nothing downstream.
- **Models are versioned data, not wiring.** Every chunk records which embedding model produced it; upgrades run blue/green as a parallel index and cut over only when evaluation approves. The reranker and the LLM sit behind their own interfaces, equally swappable.
- **Dependencies point one way.** Sources are permanent; derivatives derive from sources; chunks, indexes, graph, and wiki derive from derivatives. Nothing downstream ever writes upstream. Water flows downhill only.

The same discipline reaches into the code itself, because a module with beautiful external contracts can still rot from within: one 3,000-line file, functions that scroll like film credits, and not a word of explanation anywhere. That's the dumplings fusing again, one level down:

- **Small files:** nothing longer than **800 lines**. A file past that is holding more than one responsibility and should be split. It's not a file anymore, it's a hostage situation.
- **Small functions:** nothing longer than **50 lines**. A function that doesn't fit on one screen is several functions in a trench coat.
- **Docs at every level:** each file opens with a doc block stating its purpose; every function and class carries its own. Code explains *how*; docs explain *what for*.
- **Directories as modules:** code splits into subdirectories along module boundaries, and **each directory has its own README.md** stating what it owns, its public API, what it depends on. A newcomer should be able to parachute into any directory and know where they've landed.

The test for every boundary, architectural or code-level, is blunt: **can this piece be changed, replaced, or removed without understanding the whole system?** If not, the seam is in the wrong place. Simplicity keeps the number of parts small; modularity keeps each part replaceable. This design insists on both.

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## The Agent That Remembers

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Watch a credit officer use any search tool for ten minutes and the same shape appears every time. The first question is never the real one.

*"What's our exposure to European commercial real estate?" produces a number. Then the actual work starts. "Only the funds, not the direct loans." "As of year end." "Now split it by country." "Does that include the Meridian facility?"*

That isn't five questions. It's one question, being sharpened.

A system that treats them as five unrelated queries hands the context back to the human every single turn, who then retypes "European commercial real estate, funds only, as of 31 December 2024" into every follow-up until they give up and go ask a colleague instead. The retrieval can be excellent and the product still useless. Search that forgets is a filing cabinet with a search box. It is not an assistant.

**So the third guiding principle, alongside simplicity and modularity, is AI: an agent is a component of this system, not a feature layered on top of it.** Agents are already load-bearing in half the sections of this document, and it's worth naming them in one place: the conformance reviewer that gates every merge, the judge model that samples production answers for groundedness, the helper that clusters failure traces into rubric items, the classifier that tags PII and MNPI at ingestion, the admin chat that explains why last night's gate tripped, the assistant that vibe-codes a steward's review screen, and the user chat itself. Even this document was drafted with an agent whose knowledge base held the requirements and whose memory held the options already rejected, which is the only reason round nineteen didn't re-propose the vector database ruled out in round two.

Every one of those agents is built from the same three parts, and the design treats all three as first-class:

- **A knowledge base.** The agent reads through the retrieval API, entitlement-filtered, and answers with citations or not at all. No agent gets a private tunnel to the database and none of them answers from what the model absorbed about the world in training.
- **Memory.** What was asked, what was answered, what the user meant by "Meridian," what they already told us they don't care about.
- **Tools behind contracts.** The same documented, audited APIs humans use, with the same ACLs and the same audit log. An agent is a client of the system, never an exception to it.

**What the user chat remembers** comes in two layers.

**Thread memory** holds the live conversation: the entities in play, the filters applied, the as-of date, the documents already retrieved and shown. Each follow-up is rewritten into a standalone question before anything else happens, and the rewritten question then enters exactly the same normalizer described in "Three Ways to Find a Needle." One code path, still. The rewrite is displayed above the answer and can be edited, so the user always sees the question the system actually asked. Guessing is allowed; guessing invisibly is not.

**Profile memory** persists across sessions: the collections this person lives in, whether they usually want in-force or historical text, their preferred answer length, their disambiguations ("Meridian" means Meridian Holdings SA, not the Meridian Fund), and the answers they previously flagged as wrong. Over a few weeks the system stops asking a credit officer whether they meant the lending collection.

Five rules keep memory from becoming a liability of its own:

1. **Memory belongs to one identity and is never shared.** It is stored per user and joined against the ACL tables on read, exactly like chunks are, before anything is ranked or shown. A remembered document ID can therefore never outlive the permission that produced it: revoke access at 9 a.m. and the 9:05 follow-up cannot see it either. Memory is not a back door with a friendly interface.
2. **Memory is context, never evidence.** It steers retrieval and resolves pronouns. It does not supply facts. Every claim in every answer still cites a chunk retrieved in the current turn, so a number that exists only in a remembered answer never makes it into a new one. When memory and the corpus disagree, the corpus wins without a debate.
3. **Memory is visible and erasable.** A panel in the user chat shows exactly what is remembered, item by item, with a button to delete any of it, a button to forget the whole thread, and a button to forget everything. Nobody has to guess what the system thinks it knows about them.
4. **Memory expires.** Thread memory lives 30 days past the last message; profile memory expires 12 months after its last confirmation, both configurable by policy. Preferences from two years ago aren't preferences, they're archaeology.
5. **Memory is audited and disposable.** Which memory items entered a prompt is part of the query trace, alongside the chunk IDs and the model versions, because "why did it answer that?" must stay answerable years later. And the tables are plain PostgreSQL, dropped and forgotten at any time: losing them costs convenience, never correctness.

The golden suite grows two new families of cases for all this: **multi-turn refinement** (does the fourth turn still answer the question the first turn was really asking?) and **adversarial inheritance** (a follow-up that tries to inherit access from an earlier turn, or from another user's session, and must come back empty). Permission leaks through a conversation are still permission leaks, and the tolerance is the same number as everywhere else in this document: 0.00%.

The admin chat gets the same machinery pointed at operations: it remembers the incident it's working, so "and what about the day before?" means something. The maintainers' workbench remembers too, which is why a monkey can carry the trace that created it.

**And the agent belongs to everybody, not just to the people who run the system.** A knowledge base that only answers questions is a library. This one is also a workshop, and the door is open to every user and every maintainer, who can ask it in plain language to:

- **build a mini-tool:** a review screen, a saved view, a one-page dashboard over the questions this team keeps asking
- **compose and schedule a job or a workflow:** extraction, cleaning, transformation, loading, verification, assembled from the same idempotent steps and run through the same transactional queue as everything else
- **run an analysis:** count, group, compare, chart, across whatever the asker is entitled to see and nothing else
- **produce a report or a document:** the quarterly summary, the memo, the pack for Thursday's meeting, with its citations attached and checkable

The split between the two audiences is one of scope, not of privilege. Maintainers get the data-plane jobs (backfills, re-parses, ETL workflows) because that is their job. End users get read-only artifacts, per requirement S6, since v1 executes no transactions and makes no decisions. Nobody gets a tool that can see more than its author can.

That last clause is the whole guardrail, and it is worth being blunt about: **everything the agent generates reaches data only through the documented, audited APIs.** So the ACLs apply automatically, the audit log fills up automatically, and a generated page cannot show its author a document they were never entitled to open. Generated code passes the same CI as human-written code: the tests, the size limits, the AI conformance review ("Monkey Business" has the mechanics). Fast where speed is cheap, gated where mistakes are expensive.

Memory is what turns a search box into a conversation. The audit trail is what keeps that conversation admissible.

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## Where Do Three Million Files Live?

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Databases make terrible file cabinets, and cloud object storage is off the table, because the compliance boundary says everything stays on infrastructure the firm controls. So originals live on **mounted volumes** (AWS EBS behind an NFS export, or the on-prem NAS), and the database stores only a `storage_uri` pointing at each file. The database knows *where* everything is; it just refuses to carry it.

That sounds mundane until you ask what three million files do to a filesystem. Dump them into one directory and every listing crawls. That's how you make a filesystem cry. Let people overwrite files in place, and you can never again prove what a citation pointed to.

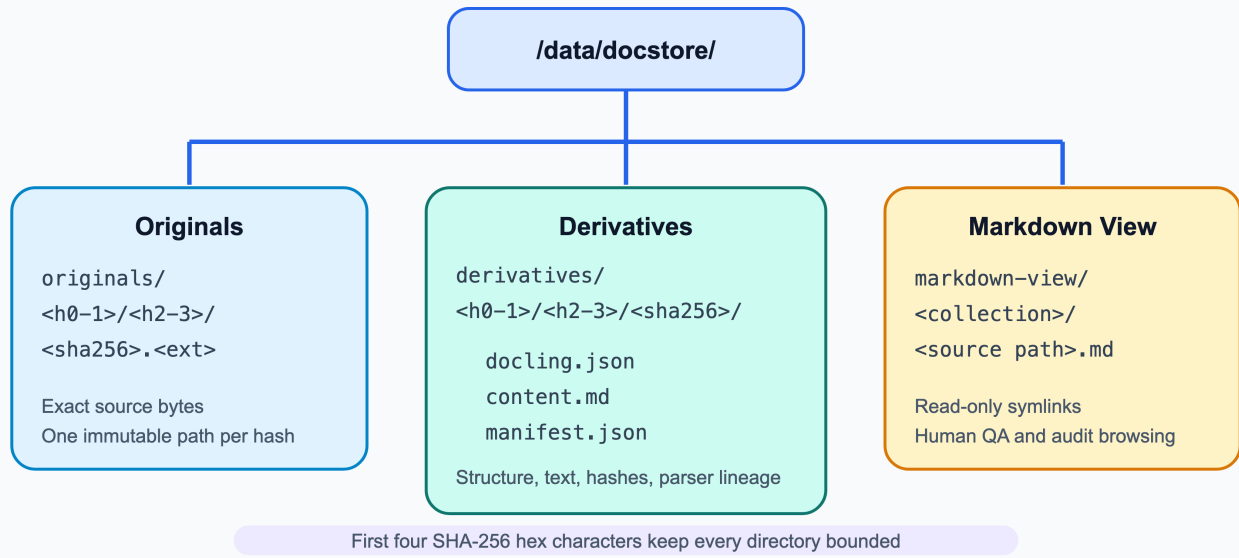
Two old-fashioned disciplines solve both:

- **Content addressing.** Every file is stored under its SHA-256 hash, sharded into subdirectories by the first four hex characters (`originals/ab/cd/<sha256>.pdf`), so no directory ever holds more than a few thousand entries. Same bytes, same path, forever; a changed document gets a *new* hash and a *new* path.
- **Write-once.** Workers write to a temp path, `fsync`, then atomically `rename()` into place. Nothing is ever overwritten. Once written, a file is immutable, optionally sealed with `chattr +i`, so even a root process can't quietly rewrite history.

This buys something quietly wonderful: **a snapshot of the volume is consistent by construction**, at any instant, no coordination required. A half-written file exists only under a temp path that nothing references. File that fact away; it becomes the hero of "The Day Everything Goes Wrong."

Alongside each original live its **derivatives**: the parsed, normalized representations (structured JSON plus clean Markdown). And because storage can lie slowly and silently, a weekly scrub job re-hashes samples and compares them against the database, hunting bit rot.

## Document Volume - Content-Addressed and Sharded



Capacity is comfortable: ~3 TB of originals, 1-2 TB of derivatives; provision 8-10 TB and grow with LVM.

## Teaching the System to Read

The corpus is a museum of formats: born-digital PDFs, Word contracts, PowerPoint decks where the real story hides in the speaker notes, spreadsheets with opinions, and scans of paper signed before some of our analysts were born. Run everything through a heavyweight OCR pipeline and the backfill takes months. Run everything through a fast text extractor and it butchers the scans and shreds every table.

**The solution is triage.** A fast path (PyMuPDF) inhales born-digital files at 50-100 pages per second per node. A heavy path, GPU workers running Docling with layout analysis and OCR, takes the hard cases at 8-15 pages per second: scans, brutal multi-column layouts, decks. Every scanned page gets an **OCR confidence score** that follows its text all the way into answers: a citation built on shaky OCR admits it, right there in the citation.

Two principles govern the pipeline:

**Parse once, index many times.** Parsing three million files costs weeks of compute, and we refuse to pay twice. Parsed derivatives are stored permanently. Re-chunking, re-embedding, upgrading models, even defecting to a different search engine: all of it reads from derivatives, never from raw files. This is precisely what makes the scaling gates cheap to cross: the entire search layer is *disposable*.

**Deduplicate before indexing.** The same memo lives in five SharePoint sites and eleven inboxes, because that is how offices work. Exact duplicates get caught by hash before they're even fetched twice; near-duplicates by MinHash/SimHash on normalized text. Aliases map every copy to one canonical version. The arithmetic is satisfying: ~45 million potential chunks collapse to 20-25 million indexed ones, half the index we'd otherwise be paying to search.

The whole pipeline runs as idempotent steps in a transactional job queue: plain PostgreSQL rows claimed with `SKIP LOCKED`, no message broker in sight. A crashed worker resumes exactly where it stopped. A re-delivered event is a shrug and a no-op.

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## The Art of Breaking Documents Apart

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Search engines don't retrieve documents; they retrieve *chunks*. Chunk badly and everything downstream suffers. Split a contract mid-sentence every 500 tokens and you get fragments that match queries but can't support answers. Treat a financial table as prose and its meaning evaporates. A number without its header row isn't data, it's confetti.

**The solution: cut along the document's own seams.**

- **Narrative documents** split at heading boundaries, 400-800 tokens, on clean paragraph breaks. Each chunk carries its full heading path (`Credit Agreement > Section 4.2 > Representations`) so it never forgets where it came from.
- **Slides**: one slide per chunk, with deck title, slide title, and presenter notes together. (In decks, the notes are where people write what they actually think.)
- **Tables**: kept whole, or split as row-groups that each repeat the header row; the raw structured version rides along in metadata for exact numeric lookups.
- **Scans**: chunked like narrative, wearing their OCR confidence; anything below 85% gets flagged in prompts and citations.

One principle rules the data model: **documents are not chunks**. A document has lifecycle, lineage, entitlements, and legal retention. It's a citizen. A chunk is a disposable search artifact, rebuildable at any time from derivatives. Confusing the two is how systems end up unable to delete what the law requires them to delete. That is an awkward conversation nobody wants to have.

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## Three Ways to Find a Needle

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Now the heart of the matter. Users ask two fundamentally different kinds of questions, and each one breaks the search method built for the other:

- *"What is our aggregate exposure to commercial real estate in Europe?"* is thematic and conceptual. Keyword search is useless; no document contains that sentence.
- *"Find clause 7.3(b) of the Meridian facility agreement."* is an exact needle. Semantic search is actively dangerous here: it will happily return something *similar* to clause 7.3(b), delivered with total confidence. Similar is worse than nothing.

**Solution zero: make both sides speak the same language.** There's a quiet problem underneath vector search. A question and a contract paragraph are different species: *"What's our CRE exposure in Europe?"* is eight words of informal shorthand; the text that answers it is three hundred words of dense legalese. Embed

them as-is and you're not comparing apples to apples; you're comparing apples to legal oranges. So both sides get pre-processed into comparable form *before* any embedding happens:

- **One normalizer, two callers.** A single shared module cleans and canonicalizes text: Unicode and whitespace normalization, acronym expansion from a curated dictionary ("CRE" → "commercial real estate (CRE)"), canonical entity names from the entity index, dates written out in full. It runs on chunks at ingestion and on queries at search time, the *same code path* by design, so the two sides can never drift apart. The dictionary is versioned like the embedding model: extend it, and the affected chunks re-embed through the usual blue/green machinery.
- **Model-specific formatting.** The embedding models in this family (BGE/E5) are trained *asymmetrically*: queries expect a `query: prefix`, passages a `passage: prefix`. It looks like a trivial string constant, and skipping it silently costs several points of recall. The most expensive missing prefix in the business.
- **Chunks arrive dressed for the comparison.** Every chunk already carries its heading path (`Credit Agreement > Section 4.2 > ...`) prepended into the embedded text, so a bare clause fragment or table row brings its context along instead of floating in space.
- **Hard queries get a stunt double.** For broad, abstract questions, an optional **HyDE** step (hypothetical document embeddings) has the LLM draft a short *hypothetical answer* in document style and embeds that instead: a fake document to find the real ones, because a fake document resembles a real document far more than a question ever will. Like the LLM reranker, it's a conditional escalation: adopted only where the golden suite shows it earns its latency.

None of this is assumed to help. Every pre-processing step is toggled and measured against the golden suite, because "common practice" is a hypothesis, not a result.

**Solution one: run both searches, always.** Semantic search (vector embeddings, computed by self-hosted models on our own GPUs) catches meaning and paraphrase. Keyword search (BM25 full-text) catches tickers, ISINs, clause numbers, party names, the stuff lawyers actually type. Results merge by reciprocal rank fusion, and a cross-encoder reranker picks the best handful.

**Solution two: search at two altitudes.** Twenty-five million chunks is a noisy haystack for a broad question. So every document also gets a one-paragraph **summary with its own embedding**: a three-million-card catalog. Broad queries consult the catalog first, shortlist the promising documents, then search chunks only within them. Needle queries, anything with an exact identifier, skip the catalog and hit the full chunk index directly, because a summary might not mention clause 7.3(b), and a missed clause is a failed audit.

**Solution three: after matching, widen the lens.** Chunks are sized for *matching*, but the sentence that matches a query and the sentence that answers it are frequently next-door neighbors. So each selected chunk gets expanded to its parent section (its siblings by heading path, within a token budget) before the model reads it. The match finds the spot; the section supplies the meaning.

**Solution four: let the agent go back for more.** Some questions cannot be answered by any single search, however clever the routing ("Why Not Just Paste It All Into the Model?"). "*Which amendments changed the leverage covenant, and what did it say before?*" needs a first pass to find the amendments and a second to fetch what they amended. So retrieval is a **bounded loop**, not a single shot: the agent reads what came back, decides whether the evidence actually answers the question, and may search again with a sharper query.

The bounds are what keep this from becoming a research project that bills by the hour:

- **At most four passes**, then the loop stops whether it is satisfied or not.

- **Every pass runs the full path**, entitlement filter included. No pass inherits access from an earlier one, and a second search can never see what the first was not allowed to see.
- **Every pass is logged** in the query trace with its query, filters, chunks, and scores, so a regulator can watch the system think.
- **Running out of passes is an abstention, with a reason.** The system says what it was still missing rather than answering from the three-quarters of an answer it managed to collect.
- **Escalation is conditional, like everything else here.** Simple lookups take one pass and stay fast; the loop switches on where the golden suite shows it earns its latency.

**Solution five: in a long document, open the table of contents.** A 300-page credit agreement or a 10-K is not a bag of chunks. An analyst handed one doesn't read random paragraphs that sound like the question; they open the contents, go to Article 7, then to Events of Default, then to (b). Similarity is not relevance: the clause that answers *"what happens if the borrower misses a payment?"* rarely looks like that question.

The open-source **PageIndex** project (VectifyAI) builds retrieval on exactly that habit, with no vectors and no chunking. Each document becomes a tree, a smart table of contents: every node holds a section title, its page range, and a short summary. At query time an LLM reads the tree and reasons its way down, branch by branch, to the sections that answer the question. Mafin 2.5, a system built on PageIndex, reports **98.7% on FinanceBench**, against roughly 50% for plain vector RAG on the same benchmark. That is a vendor's number on a public benchmark, not ours, but the idea fits our corpus well: long legal, financial, and technical documents with real structure.

We take the idea, not necessarily the library:

- **The tree is one more derivative.** Docling already recovers the heading hierarchy (requirement F3), so the tree is built from parsed derivatives, never from raw files. Node summaries are generated once per document version and stored in a plain PostgreSQL table (`doc_tree_nodes`: document version, parent, title, page range, summary). Like chunks, it is a disposable projection, rebuilt at will.
- **Only where it pays.** Trees are built for long, well-structured documents (agreements, prospectuses, filings, policy manuals). A two-page memo doesn't need a table of contents.
- **It complements search, it doesn't replace it.** Tree reasoning works *inside* a document; it cannot pick one document out of three million. So the catalog and hybrid search shortlist the documents, and the tree finds the section within them. Needle queries with an exact clause number still go straight to BM25, which is faster and cheaper.
- **A descent is one pass of the bounded loop**, capped by depth and token budget, and it runs behind the same pre-ranking ACL join. Tree nodes inherit their document's entitlements; there is no side door.
- **The path is the explanation.** The route taken (`Credit Agreement > Article 7 > 7.3 Events of Default > (b)`) is logged in the query trace and maps straight to a page-level citation. An auditor can read *why* the system went where it went, which is more than a cosine score ever tells them.
- **Measured, not assumed.** Each descent costs LLM calls and latency, so tree navigation is a conditional escalation like HyDE and the LLM reranker. FinanceBench-style questions join the golden suite, and the mode switches on for the document types where it beats chunk retrieval.

The PageIndex code itself is a reference implementation to evaluate against, not a default dependency. If it wins over a few hundred lines of our own, it enters through the same 30-day, pinned, verified door as everything else.

And beneath all of it, without exception, sits the entitlement check. **ACL filtering happens in SQL, before ranking.** An unauthorized document doesn't rank low. It doesn't exist. The permission-leak tolerance is written down as a number, and the number is **0.00%**.



## The Web Between the Documents

A few weeks into any project like this, someone asks the question that vectors and keywords simply cannot answer: *"Which amendments modify this credit agreement?"*

Similarity search finds text that *resembles* other text. But an amendment doesn't resemble the agreement it modifies; it *references* it, tersely, by title and date, the way legal documents nod to each other across the room. Those relationships (supersession, amendment, exhibit, citation, same deal) are the corpus's skeleton, and text search is completely blind to it.

Here's the lucky break: legal documents *announce* their relationships. "This Amendment No. 3 to the Credit Agreement dated March 15, 2024..." is a database row wearing a costume. No AI required: regexes and parsers pluck these cross-references out during ingestion, along with the entities (parties, ISINs, deal names).

**So the design adds a document graph, in plain PostgreSQL tables:** typed edges (`amends`, `supersedes`, `exhibit_of`, `references_clause`, `cites`) between documents, plus an entity index recording who is mentioned where. Every edge remembers how it was extracted and with what confidence. No anonymous edges.

The graph earns its keep twice:

**At retrieval time**, after hybrid search returns its chunks, the engine walks one hop. Retrieve a credit agreement, and its amendments come along, even though they never matched the query. Retrieve something with a `supersedes` edge pointing at it, and the answer gets *flagged as superseded*, structurally, rather than by hoping the newer version happened to rank well. And every hop joins the ACL table first: a link must never reveal so much as the *existence* of a document the user can't see.

**For humans**, the graph renders as an **interconnected wiki**: one Markdown file per document, YAML frontmatter on top (type, dates, classification, status), `[[wikilinks]]` for every relationship. It is browsable in Obsidian, greppable with ripgrep, and laid out in a readable tree (`wiki/documents/<collection>/<type>/<year>/...`, plus a stub page per entity listing everywhere it appears). Compliance officers get to browse the corpus by its actual structure instead of playing search-term roulette.

One rule keeps the wiki honest: **it is a projection, not a source**. Nobody edits wiki files; they're rendered nightly from the database and can be deleted and regenerated without losing a byte of truth. And since files on disk have no row-level security, the full wiki renders only inside the curators' enclave. Links leak existence, and around here, existence is confidential too.

Later (v2), community detection over this graph will cluster the corpus into themes with summary pages: the corpus explaining its own neighborhoods.

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## What Was True Last March?

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Sooner or later, probably sooner, someone will ask: "*What was the covenant threshold as of Q3 2024?*"

A system that only knows "current vs. superseded" answers with today's amendment: fluently, confidently, and wrongly. In finance, *when* something was true is half the fact.

**The solution is to give time a seat at the table.** Every document version carries an effective window (`effective_from` / `effective_to`); every graph edge carries a validity window. The dates come from the documents themselves (effective-date clauses are dated boilerplate, extractable with a regex and a straight face) and from lineage: creating a supersession edge automatically closes the predecessor's window.

On top of that:

- The retrieval API accepts an **as\_of\_date**. Instead of filtering to current versions, it filters to versions *in force on that date*, and the answer is clearly flagged as historical, so nobody mistakes last year's covenant for this year's.
- When retrieved passages disagree *because they come from different eras*, the system doesn't wring its hands about a "contradiction." It lays out a **timeline**: "the threshold was X from March 2024, amended to Y effective November 2024." Not a bug; a biography.
- Ranking gets a sense of time: queries with date intent boost chunks near that date; queries without it get a gentle nudge toward in-force versions, so stale text can't out-charm live text on similarity alone.

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## The Corpus Never Sits Still

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Everything so far described a system at rest. Real corpora don't rest: new documents arrive, versions supersede, deletions happen, and, most dangerous of all, someone's access gets revoked. Each event must ripple through every layer we've built: registry, chunks, vectors, indexes, summaries, graph edges, wiki pages. Miss one layer, and the layers start lying to each other.

**The solution is a defined lifecycle with three hard rules.**

When a document is **added**, it flows through the standard pipeline and is searchable within minutes. When one is **updated**, the new version gets fresh chunks and a **supersedes** edge; the old version's chunks are *deactivated, never deleted*. That's rule one, **soft-deactivation**, because an audit-log citation from last year must still resolve to the exact text the user saw, for as long as the records policy says. When a document is **removed** at the source, it's tombstoned: invisible to retrieval the instant the transaction commits (the active-flag and ACL filters run before ranking, so there's no index rebuild to wait for), while the underlying files quietly ride out their retention period.

Rule two: **the visibility flip is atomic**. Old chunks off, new chunks on, current-flag moved, all in one transaction. No query ever catches a document half-dressed.

Rule three: **ACL changes outrank everything**. A revoked entitlement propagates within **five minutes**, monitored as a first-class alert, with a daily full re-sync to sweep up anything a missed webhook dropped. Stale content is embarrassing; stale permissions are a breach.

The slower layers refresh on their own schedule: wiki re-renders and link re-extraction nightly, cluster refresh and physical garbage collection weekly to monthly. And because deactivated vectors pile up as dead weight in the vector index, a weekly recall check compares the index against a brute-force scan on a sample; if recall sags, that partition gets rebuilt during a maintenance window. Even indexes need a gym membership.

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## Trust, but Verify

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This system changes in two ways, and both can break it without making a sound. Engineers change the **code**, and no compiler complains when someone imports the database driver into the domain logic or bypasses the retrieval API "just this once." The **corpus** changes daily, and RAG systems degrade quietly while hallucinating confidently: a connector dies on a Tuesday and nobody notices until someone asks about a document from the

gap; an embedding upgrade shuffles rankings and recall drops three points with no error message anywhere. Silence, it turns out, is not a good sign. Silence is just silence.

**So verification runs at both timescales: every code change passes a pyramid of tests before it merges, and every day's corpus changes pass an evaluation gate before the night is out.**

## Guarding the code

Rules that nothing enforces are wishes. And untested code develops a second disease: nobody dares touch it. A codebase that can't be safely changed is the fused dumpling clump's final form: frozen solid. So every change runs the gauntlet:

- **Unit tests** guard the functions: fast, isolated, no network, no database, dependencies faked at the port interfaces, which is only possible because the ports exist. Modularity pays its first dividend here.
- **Module tests** guard the contracts: each module (connectors, parsing, chunking, retrieval, graph, wiki renderer) is tested in isolation through its **public API**, dependencies stubbed. If you can't write a module's tests without reaching into another module's internals, congratulations: you've found a boundary in the wrong place, cheaply.
- **Integration tests** guard the real seams: SQL against an actual PostgreSQL in a disposable container (pinned by digest, naturally), and the pipeline end-to-end against a fixture corpus (a scanned PDF, a deck, a spreadsheet, a near-duplicate pair), confirming that ingestion, chunking, embedding, and retrieval still agree with each other.
- **Architecture-conformance tests** guard the rules themselves. The mechanical rules go to mechanical tools: `import-linter` fails the build if domain code imports infrastructure; file-length, function-length, and docstring checks run in CI, tirelessly and without opinions. But some rules need *judgment*. Is this new dependency direction legitimate? Is this "utility module" quietly becoming a junk drawer? For those, **an AI agent reviews every change against the written architecture rules**, flags violations with the specific rule it believes is broken, and gates the merge like any failing test. Humans adjudicate disputes; the agent just makes sure erosion never happens *silently*.

## Guarding the answers

Code tests prove the machine works. A separate discipline proves it's still *right*. The yardstick is a **golden test suite**: 300+ real questions curated with Legal, Risk, Credit, and Operations, with known correct answers *and known correct citations*. It includes as-of temporal cases, adversarial permission-leak attempts (single-turn and across a conversation), multi-turn refinement chains that check the fourth turn still answers what the first turn was really asking, multi-hop questions that no single retrieval pass can answer, and, the fun part, an **unanswerable set**: questions whose answers are deliberately absent, locked behind entitlements, or found only in superseded text. The correct response is abstention. A confident answer to an unanswerable question is a hallucination caught red-handed.

But where do test questions and grading criteria come from? The tempting shortcut is a conference-room checklist: "answers should be accurate, relevant, and clear, scored 1 to 10." Checklists born this way fail twice: the scores are too mushy to act on, and the criteria only cover failures somebody predicted. The system's actual failures have more imagination than that.

**So the evals are grown from real outputs, not abstract principles.** During the pilot, evaluation starts with actual **traces** (the query, the routing decision, every chunk retrieved with its scores, the final answer) put in front of domain experts in a review interface where they annotate problems *in context*: the sentence that overstates,

the citation that doesn't support its claim, the abstention that should have been an answer. **Failures get reviewed before criteria get written.** AI helps with the labor (clustering similar failures, surfacing patterns, drafting rubric items), but humans supply the taste: only Legal knows which paraphrase of a covenant is materially wrong.

From those annotated traces, criteria emerge along two paths:

- **Top-down criteria** encode what the task always required: every claim carries a citation; the abstention phrasing is exact; nothing in the answer comes from outside the retrieved context; format and policy hold.
- **Bottom-up criteria** capture what the traces actually revealed, patterns no whiteboard session would have produced: *"quotes the superseded threshold when both versions were retrieved," "merges two counterparties with similar names," "answers from general legal knowledge instead of the documents."*

Every criterion becomes a **specific pass/fail check, never a vague score**. Not "faithfulness: 7/10" but "every numeric claim appears in a cited chunk: yes or no." Binary checks are debuggable: a failure points at exactly one thing to fix. And when the checks are automated with LLM judges, **each criterion gets its own judge pass**. One mega-prompt asked to verify twelve requirements will quietly forget a few; twelve small judges with one question each forget nothing.

Then the flywheel closes: every validated failure pattern graduates into permanent machinery: a regression case in the golden suite, a CI check that runs before any prompt or model change ships, a dashboard series, a production monitor. The suite is the accumulated memory of every way the system has ever been caught being wrong. It only grows.

## The nightly gate

Every night, after the day's sync settles, the full suite runs against **production** indexes:

- Results compare against a rolling 7-day baseline. Recall down more than 2 points? Citation precision slipping? Abstention correctness off? **Any** permission leak at all? The gate trips: on-call gets paged and ingestion freezes until a human understands why. A failed eval is treated exactly like a failed deploy, because that's what it is.
- A **reconciliation count** runs across three layers (source systems → registry → active chunks), so a silently dead connector shows up as a number that doesn't match *tonight*, not as a mystery at quarter-end.
- A **self-retrieval probe** samples documents added or changed that day, generates a query from each one's own content, and verifies the document actually comes back. Proof that new content is *findable*, not merely stored somewhere warm.

And because golden questions can't cover real traffic, hallucination gets a continuous watch: a **self-hosted judge model** samples production answers daily and checks entailment. Does each cited chunk actually support the claim leaning on it? Abstention rates are watched in both directions: a sudden drop means the system got brave about things it shouldn't answer; a spike means retrieval got worse. And monthly, humans re-score a sample of the judge's verdicts, because a judge nobody audits stops being a judge.

One principle at both timescales: **a failing test blocks the merge; a failing eval freezes ingestion. The system proves it still works after every change, or the change doesn't ship.**

## The System That Heals Itself

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Detection is only half a nervous system. At this scale, something is *always* slightly broken: a worker dies mid-parse, a file quietly rots on disk, a connector hiccups and drops an event, an index puts on dead weight. If every small failure needs a human, the on-call engineer becomes the system's immune system, and humans make terrible white blood cells. They sleep, they take vacations, they burn out. Meanwhile small failures wait in line, and small failures that wait long enough grow up to become incidents.

**So the design adds a fourth principle, and it is not a lesser one: self-healing. When something goes wrong, the system cleans and repairs itself. Humans get paged as needed, for the exceptional, never for the routine.**

The repair reflexes, most of which we've already met wearing other hats:

- **A crashed worker heals by retry.** Every pipeline step is an idempotent job in the transactional queue, claimed under a five-minute lease that the worker heartbeats while it works. A dead worker stops heartbeating, the lease expires, and the next worker picks the job up. Re-running a finished step is a no-op. Nobody restarts anything by hand at 3 a.m.
- **A corrupted file heals from snapshots.** When the weekly scrub finds a file whose hash no longer matches the registry, it quarantines the file, restores the most recent snapshot copy that *does* verify, re-checks the hash, and files a report. The human reads the report over coffee, after the repair.
- **A missed event heals by reconciliation.** The nightly three-layer count doesn't just detect gaps; it queues refetch jobs for every missing document. The daily full ACL re-sync does the same for dropped permission events. Detection and repair are one motion.
- **A degraded index heals by rebuild.** When the weekly recall check finds a partition sagging under dead vectors, that partition gets re-indexed in the next maintenance window, scheduled automatically and mentioned in the morning summary.
- **Everything derived heals by regeneration.** Wiki pages, symlink trees, summaries, chunks, whole indexes: all projections of the source of truth. The universal repair tool is gloriously dumb: throw the broken thing away and rebuild it. This is where the earlier principles pay compound interest: *parse once, everything downstream disposable* means self-healing rarely requires cleverness, just a rebuild job and patience.

Self-healing has manners, though. Retries use backoff and give up after a set number of attempts. A job that keeps failing retires to a dead-letter state and pages a human, because infinite retry is not persistence, it's a tantrum. And two things are **never** repaired by automation guessing: entitlements and regulated records. Anything ambiguous in those neighborhoods escalates to people immediately. The system heals itself; it does not *improvise* itself.

### Restart Is a Non-Event

Machines get rebooted. Deploys roll. A GPU node runs out of memory at four in the morning and comes back forty seconds later, and the patching window reboots the database host on the second Tuesday of every month whether anyone is ready or not. If putting the system back together after any of that needs a human, it isn't self-healing. It's just well documented.

**So here is the rule underneath all the others: no durable state lives in process memory.** Everything a restart could destroy sits in PostgreSQL, on the write-once volume, or in the category of things we can cheerfully rebuild. A process here is a worker, never a vault.

Trace it through the things that could be in flight when the lights go out:

| Interrupted                                       | What a restart finds                                                                                                                                                  |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A parse job halfway through a 900-page scan       | A queue row whose lease has expired. The next worker claims it and starts that step again from the beginning, which is safe because every step is idempotent          |
| A backfill three weeks into a five-week run       | Registry rows and content-addressed files that already exist. Work already done is detected by hash, skipped, and never repeated                                      |
| A user mid-conversation                           | Thread memory in a table, so the conversation resumes exactly where it was, on any node, with no sticky sessions and nothing lost but the request that was in the air |
| An index build                                    | A shadow table that was never flipped into service. An interrupted build costs wasted CPU and nothing else; no half-built index ever answers a query                  |
| A visibility flip (old chunks off, new chunks on) | Either the whole transaction or none of it. Postgres has been good at this for thirty years and we are not going to improve on it                                     |
| A file being written                              | A temp path nothing references, swept on the next scrub. The atomic <code>rename()</code> either happened or didn't                                                   |

The boot sequence is scripted, because the components have to agree with each other on the way up just as they do after a restore:

1. **Mounts are verified before anything starts.** A missing volume must fail loudly at boot, not quietly at the first query, and a spot-check of a few hashes proves the right volume came back.
2. **PostgreSQL completes crash recovery** by replaying its WAL, and accepts no connections until it's finished. This is ordinary database behavior and it's the single largest reason the design keeps truth in one.
3. **Workers start and reclaim expired leases.** The queue drains from wherever it stopped.
4. **The ACL sync runs before the API answers its first question**, for the same reason it runs before the API reopens after a restore: serving stale entitlements is a breach whose root cause is a runbook.
5. **Models load and warm up**, and the readiness probe stays red until a real query returns a real answer. A cold GPU node that accepts traffic is a node that times out.
6. **Reconciliation counts run once service is up**, because the outside world kept changing while the system was down.

Two things are deliberately *not* resumed: in-flight HTTP requests and in-flight multi-step retrieval loops. Neither has written anything, both are cheap to ask again, and a system that tries to resurrect a half-finished conversation turn is a system inventing state it never persisted. The client re-asks. That's the whole recovery procedure.

And it gets exercised rather than assumed. Every deploy is a rolling worker restart, so the common path is proven several times a week by the ordinary business of shipping. The full-stack restart, database host included, is rehearsed in the quarterly drill alongside the restore test, on a stopwatch, and what came back and how long it took goes into the morning report.

The quiet payoff: the team's time goes into making the system better, not holding it together. A platform that needs constant human attention isn't simple, no matter how few boxes are on its diagram.

## Answers You Can Take to a Regulator

The generation step, the part everyone points at and calls "the AI," is deliberately the most constrained component in the entire system. It has the least freedom of anything described in this document, and that's a compliment.

The model receives only the top-ranked, entitlement-filtered, section-expanded chunks, each wearing a name tag: document, page, version. The contract is strict:

- **Every factual claim carries a citation token**, validated by middleware against the chunks actually retrieved. An answer with an unverifiable citation is rejected before any user sees it. The model cannot cite what it was never given; the bibliography is closed-book.
- **Contradictions get surfaced, not smoothed over**, and when versions differ across time, they're presented as a timeline ("What Was True Last March?").
- **Insufficient evidence gets an honest refusal**: *"The available documentation does not contain sufficient evidence to answer this inquiry."* The system is graded on saying this at the right moments, which is exactly what the unanswerable set is for. Saying "I don't know" well is a skill, and here it's a tested one.

For questions where being wrong is expensive, the design offers two stronger modes, both built on the same insight: **hallucinations are idiosyncratic, but truths converge**. Two models rarely invent the *same* falsehood; they routinely agree on the same fact.

- **Fusion mode**. Two or three *independent* self-hosted models answer the same question from the same retrieved context, and a judge model synthesizes: claims that multiple models make, with citations that check out, survive; claims that only one model produces get treated as probable hallucinations and dropped or flagged. It's the jury system for language models: no single juror's imagination becomes the verdict. The cost is honest (3-4× inference), so fusion is tiered rather than default. It switches on for sensitive collections, for queries where retrieval confidence is low, and on demand, through a "verify mode" toggle in the user chat for the answer someone is about to put in front of a client.
- **Persona cross-examination**. The cheaper sibling: one model, several perspectives. The same question is answered as a credit officer, then reviewed as a skeptical auditor, then as opposing counsel looking for the weak spot. Where the personas agree, confidence rises; where they diverge, the answer says so, and divergence itself is a signal worth surfacing, because a claim that survives cross-examination from three angles is a claim standing on evidence rather than on style.

Both modes ship behind the usual gate: measured on the golden suite and the judged groundedness rate, adopted where they demonstrably reduce hallucinations within the latency budget, and skipped where a single well-cited pass already earns its keep. More jurors are only worth paying when the verdict is in doubt.

And everything is written down. Every query logs the user, their evaluated roles, the question as typed and as rewritten from conversation memory, the memory items that entered the prompt, the applied filters, every chunk ID retrieved with its scores, the model and prompt versions, the generated answer, and the rendered citations. Monthly partitions of this log are exported nightly to **write-once storage** with hash manifests, including the chunk *text* itself, so the record stays self-contained even after some future re-chunking retires the IDs. Deletion happens through the records-management process, on the retention schedule. Not through engineering. Ever.

## Three Doors for Humans

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So far this design has been very generous to machines (APIs, queues, contracts, agents everywhere) and has offered humans exactly one amenity: a wiki. That won't do. A system without windows gets operated by SSH and psql, which is a fancy way of saying "an incident waiting for a typo." And a Q&A system without a place to ask questions is a philosophical exercise.

**So the design includes one web frontend with three doors, one for each kind of human.**

**Door one: the command center.** Dashboards and controls for the people who run the system:

- **Watch:** connector health and ingestion queues, index lag against the propagation SLAs, the nightly gate's trend lines (recall, citation precision, abstention correctness, groundedness), reconciliation deltas, self-healing activity (what got retried, restored, rebuilt overnight: the morning-coffee report).
- **Control:** start or pause data loads and backfills, trigger re-indexing or re-parsing for a collection, manage removals and tombstones, kick off validation runs and golden-suite tests on demand.
- **With manners:** every button drives the same documented, audited APIs used everywhere else; the command center holds no private tunnel to the database. Destructive actions demand confirmation, and everything lands in the audit log with a user identity attached. The dashboard is a *view* with steering, not a backdoor.

**Door two: the admin chat.** An agentic chat for operations: the command center's conversational twin, which remembers the incident it is working on so a follow-up doesn't restart the investigation. "Why did last night's gate trip?" "How far behind is the SharePoint connector?" "Re-run reconciliation for the lending collection." The agent answers by calling the same admin APIs the dashboards read, so it can *explain* as well as display. Connecting an eval regression to the embedding model that shipped the day before is exactly the kind of dot-connecting agents are good at. Read operations flow freely; anything that changes state requires an explicit confirmation step; anything touching entitlements or records stays human-only, per the self-healing rules. Every agent action is audited like a human one.

**Door three: the user chat.** The main event, where lawyers, credit officers, and analysts actually meet the system. Ask a question, get a cited answer. Citations are clickable and open the exact page of the exact version in a document viewer. An **as-of date picker** turns time-travel queries from a power feature into a dropdown. The chat **remembers the conversation and the person having it**, so follow-ups refine instead of restarting, with the rewritten question shown above each answer and a memory panel listing everything retained, item by item, each with a delete button ("The Agent That Remembers"). Abstentions are displayed honestly, as answers, not apologies. Users can also put the agent to work rather than just questioning it: ask it for an analysis, a report, a document, or a small saved view over what they are entitled to see, all of it read-only in v1 and all of it through the audited APIs. And there's a **flag button** on every answer: one click sends the full trace into the evaluation review queue and gives the user a monkey they can follow to its resolution, which means every dissatisfied user quietly contributes to the golden suite. The complaint department feeds the immune system.

**The frontend follows the house rules.** It's built in **vanilla JavaScript**: no React, no frameworks, no build step. The browser already ships a perfectly good runtime; for three pages of dashboards and two chat panels, a framework is another dependency to quarantine, another supply chain to audit, and another migration in five years when it falls out of fashion. The code is **modular**: small ES modules, one per feature (chat panel, dashboard tiles, document viewer, citation renderer), each obeying the same limits as the backend: files under

800 lines, functions under 50, docs at every level, a README per directory. And **all styles live in one `styles.css`**, one file to open and one place to look, with no styles hiding in JavaScript. Plain static files served from the same infrastructure, talking only to the documented APIs. The simplest frontend that fully works, which is, by now, the house style.

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## Monkey Business

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The corpus isn't maintained by "the system." It's maintained by a group of humans: data stewards who re-scan bad batches, review low-confidence OCR, resolve deduplication conflicts, approve entity merges, and chase down the connector that's been sulking since Thursday. That work needs owners, statuses, and handoffs. Otherwise it lives in email threads and in the heads of people who might be on a beach next week.

**So the design gives maintainers a task system, and the tasks are called monkeys.** A monkey is the next move: the task, the ticket, the thing on your back that needs to hop off. *"Re-parse the Q3 scans."* *"Review 40 chunks below OCR threshold."* *"Two documents claim to be the canonical version. Pick one."* Each monkey knows its owner, status, priority, and history, and carries links to the actual artifacts: document IDs, job IDs, the failed eval case, the flagged answer.

The elegant part is where monkeys come from. The system already produces them; it just didn't have a word for them yet:

- Self-healing gives up on a job after its retries → the dead-letter entry **becomes a monkey**, assigned to the right group.
- The nightly gate trips → a monkey, with the regression report attached.
- A user flags a bad answer → a monkey in the review queue, trace included.
- The scrub job finds a file it can't restore, the reconciliation finds a document the connector can't refetch → monkeys.
- And humans create their own: curation drives, collection onboarding, cleanup campaigns.

Every escalation path in this document now has an inbox, an owner, and a status. The immune system files tickets.

**The human mechanics are deliberately ordinary.** Everyone sees *their own* monkeys: what's open, what's blocked, what failed overnight. They can act on them, hand them to someone else (vacation transfers a whole troop in one action), or **escalate** them up to a group or a manager when the answer is above their pay grade, which is a normal Tuesday and not an admission of defeat. Managers see everything: all the monkeys, their status and age, who's drowning and who's idle, because a task system where managers can't see the pileup is just a diary.

The end users get a narrow window on the same system rather than a separate one: the analyst who flagged a bad answer can see that their flag became a monkey, who owns it, and how it was resolved. Complaining into a void teaches people to stop complaining, and the complaint department is how the immune system gets fed.

**No failure is orphaned.** Every escalation path in this design terminates in a monkey with an owner, a status, and a history, and a monkey with no owner is itself an alert.

**Architecturally, this is a separate FastAPI web app**, its own module with its own API and its own vanilla-JS frontend (house rules: ES modules, one `styles.css`, no frameworks). Separate on purpose: the retrieval

service stays lean and stable while the maintainers' workbench evolves at business speed. The monkeys themselves live in plain PostgreSQL tables (one system of record, as always) and the app talks to the rest of the platform only through the documented APIs.

And because the workbench evolves at business speed, **maintainers get the full-strength version of the agent's workshop** ("The Agent That Remembers"), including the data-plane jobs that end users don't get. A steward who needs a review screen for last month's low-confidence scans, or a one-page dashboard for a cleanup campaign, describes it and lets an AI assistant generate it: a small vanilla-JS page over the existing APIs, shipped in an afternoon, no platform-team ticket required. The same goes for **workflows**: maintainers create and schedule data jobs (extraction, cleaning, transformation, loading, verification) composed from idempotent steps and run through the same transactional job queue as everything else.

Vibe-coding gets guardrails rather than a leash. Generated tools and workflows live inside the maintainers' app, consume only the documented APIs (so ACLs and audit apply automatically; a vibe-coded page cannot see what its author can't), and pass the same CI as human-written code: the tests, the size limits, and the AI architecture-conformance review. Fast where speed is cheap, gated where mistakes are expensive. The business gets its tools in hours; the platform keeps its seams.

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## The Day Everything Goes Wrong

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Now the section nobody enjoys writing. A reboot isn't a disaster and doesn't belong here; that path is a non-event and lives in "The System That Heals Itself." This section is about the day the data itself is in danger. Suppose the worst, all of it, preferably on a Friday: a bad ACL sync poisons entitlements, an operator fat-fingers a `DROP TABLE`, ransomware arrives, an availability zone catches fire. What survives?

First, an unpopular truth: **replicas are not backups**. A standby replica will faithfully replay your `DROP TABLE` within milliseconds. That's its job, and it's very good at it. Replication protects against hardware failure. Backups protect against mistakes and malice, which have no SLA.

So: the database ships every write to an archive continuously (point-in-time recovery to any second; losing at most one hour is the written objective), plus weekly full and daily incremental base backups. Backup storage lives in a **separate account with independent credentials**, so no single compromised admin can torch both the data and its safety net in one evening. Volume snapshots, consistent by construction thanks to the write-once discipline, are copied cross-AZ and **locked** against deletion: even ransomware holding admin keys can't erase the past. The audit archive is write-once by design.

Restores follow a scripted order, because the database and the volume must agree with each other afterward: **database first**, to the chosen moment; **volume snapshot from at-or-after** that moment (write-once means a newer snapshot only carries harmless extras, while an older one might be missing referenced files; newer is always safe, older never is); then a reconciliation pass that re-hashes every referenced file and queues re-fetches for gaps. **ACLs re-sync before the API reopens**. Restoring stale entitlements would be a permission leak with a runbook as the root cause, which is not a sentence anyone wants in a postmortem.

And because a backup that's never been restored is a rumor: **a monthly automated restore test** rebuilds the database in an isolated environment, verifies integrity, and runs the golden suite against it. A failed restore pages on-call like a production outage. Quarterly, a full disaster-recovery drill runs the whole sequence against

a stopwatch, together with a full-stack restart of the live system, and the results go to Compliance and Business Continuity in writing.

Recovery targets, stated plainly: lose at most one hour of data; be back within eight hours for a full-site event, faster for lesser disasters.

## How Big Is This, Really?

The numbers that make the single-database bet rational rather than reckless:

### On the document volume:

| What                                 | Size           |
|--------------------------------------|----------------|
| Originals (3M × ~1 MB)               | ~3 TB          |
| Parsed derivatives (JSON + Markdown) | ~1-2 TB        |
| Rendered wiki tree                   | ~0.3-0.5 TB    |
| Headroom for growth                  | ~2-3 TB        |
| <b>Provisioned</b>                   | <b>8-10 TB</b> |

### In PostgreSQL:

| What                                  | Size               |
|---------------------------------------|--------------------|
| Chunk text (20-25M chunks)            | ~40-60 GB          |
| Vectors + HNSW index (half-precision) | ~50 GB             |
| Full-text (BM25/GIN) indexes          | ~20-30 GB          |
| Registry, ACLs, summaries             | ~15-25 GB          |
| Document graph                        | ~5-10 GB           |
| Agent memory (threads + profiles)     | ~2-5 GB            |
| Audit log growth                      | ~2-5 GB/month      |
| <b>Steady state</b>                   | <b>~200-300 GB</b> |

A single instance with 128-256 GB of RAM keeps the vectors and hot indexes in memory, where they belong. Backups run 2-3× database size; the first volume snapshot equals used bytes, with small daily deltas after.

Speed targets: **p95 under 1.5 seconds** for a retrieval pass, under 4.5 end-to-end for the single-pass answer that most questions get. Queries that escalate to the multi-step loop carry their own budget, **p95 under 10 seconds** for four passes, with each pass shown to the user as it happens, because a system that is visibly working is easier to wait for than a spinner. Ingestion: the full backfill lands in weeks, bulk-loaded into unindexed tables with the indexes built once at the end, because building indexes *during* a 25-million-row load is how a backfill becomes a quarter-long hostage negotiation. After that, incremental sync in minutes.

## Don't Chase the Latest Version

One attack vector remains, and it's the one most teams leave wide open, enthusiastically, on purpose, every day.

The whole design keeps data inside the firm's walls: self-hosted models, mounted volumes, private endpoints. And then a developer types `pip install`, and the build cheerfully reaches out to a public registry and downloads whatever was uploaded yesterday. By anyone. Modern supply-chain attacks live exactly there: a hijacked maintainer account ships a poisoned point-release; a typosquatted package waits patiently for one mistyped name; a backdoor hides in a build script. The 2024 `xz` incident came within weeks of shipping inside every major Linux distribution. The freshest package is the least reviewed package. "Latest" is not a version. It's a gamble.

**The solution: every dependency goes through quarantine, pinned, aged, and verified.** Like new hires, packages need a probation period.

**Python itself** comes from Homebrew (macOS or Linux), pinned to a major.minor line and upgraded on purpose, never by accident:

```
brew install python@3.12
```

**Environments and modules** are managed with `uv`, under two standing rules: nothing published in the last 30 days, and no prereleases. Let the rest of the world do the beta testing; they volunteer daily. In `pyproject.toml`:

```
[tool.uv]
exclude-newer = "30 days"      # ignore anything published in the last month
prerelease = "disallow"       # never alphas, betas, or release candidates
```

Upgrades become an explicit, reviewable act:

```
uv lock --upgrade
uv sync
```

Or, for a `requirements.txt`-based flow:

```
uv pip install --upgrade --exclude-newer "30 days" -r requirements.txt
uv pip install --reinstall --exclude-newer "30 days" -r requirements.txt
```

The lockfile is committed, so every environment (laptop, CI, production worker) resolves to byte-identical, properly-aged packages. No surprises, which is the highest compliment in operations.

**PostgreSQL follows the same rule.** Development machines run a pinned major line from Homebrew (`brew install postgresql@16`); production uses the official PGDG repositories with the version held explicitly. Minor releases get adopted a few weeks after they ship, once the ecosystem has kicked the tires. A brand-new major, the dreaded `.0`, never goes straight to production: wait for the first minors, and rehearse the upgrade on the monthly restore-test instance, which conveniently exists anyway.

**Docker images get the strictest treatment**, because a tag is a promise nobody has to keep:

- Official images only ( `postgres` , `python` ), never the look-alikes.
- Pinned **by digest** rather than by tag ( `postgres:16.6@sha256:...` ), so the image cannot change underneath you. `:latest` is banned outright.
- Pulled once into the firm's **internal registry**, scanned (Trivy/Grype) on the way in, and only then available to builds. Production hosts pull from the internal registry, never from the public one.

The system guarding three million confidential documents should not be one `pip install` away from running last night's upload. So it isn't.

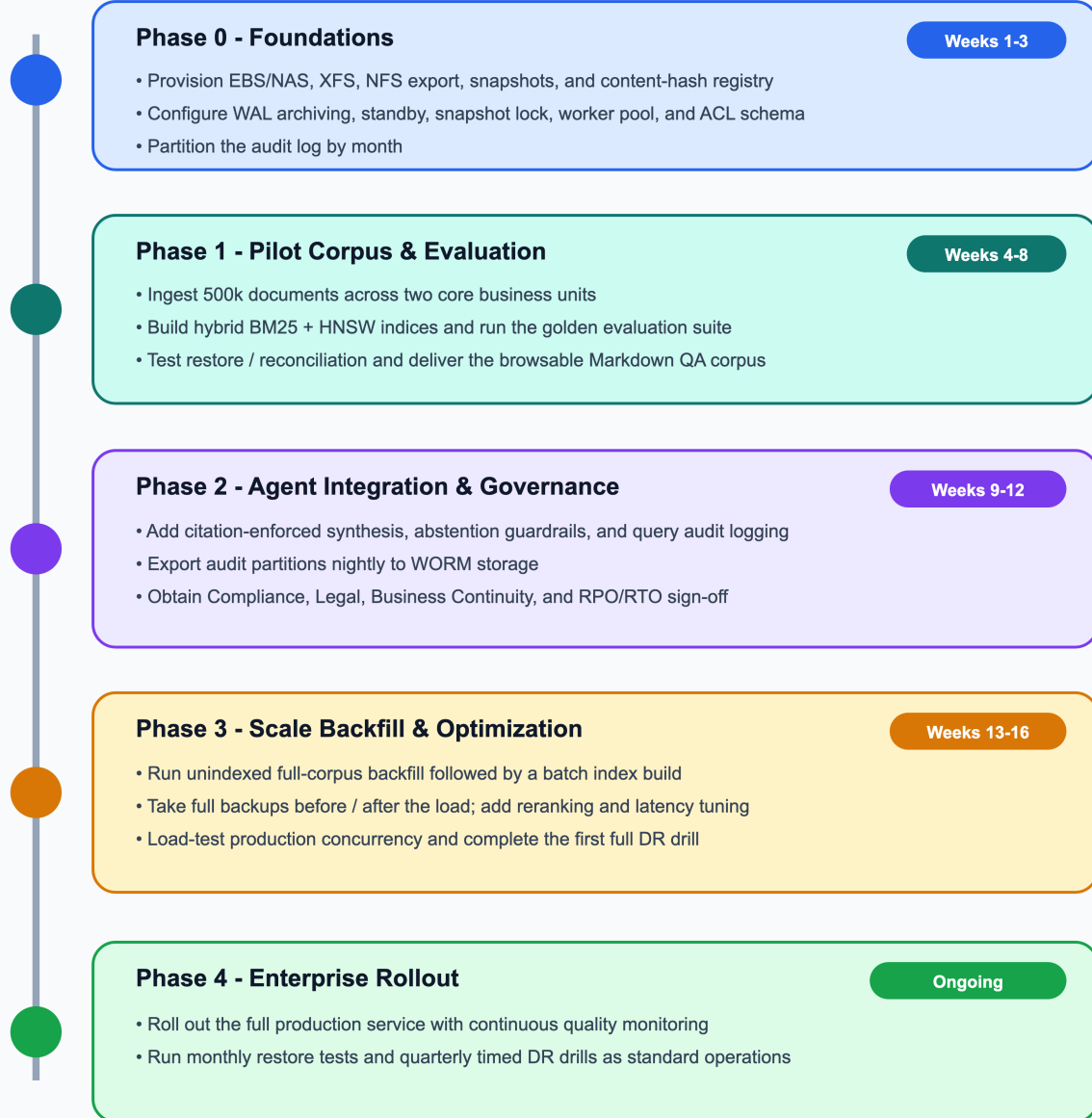
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## The Road from Here

---

The build order is chosen so that trust is earned before scale is attempted. Prove it, then grow it:

## Delivery Roadmap



**Phase 0 (weeks 1-3)** lays the foundations: storage, database, WAL archiving, ACL schema. **Phase 1 (weeks 4-8)** ingests a 500k-document pilot across two business units, builds the hybrid indexes, extracts links and effective dates, renders the first wiki, and runs the golden suite for the first time, while curators grade the extraction on the same corpus they're QA-ing anyway. **Phase 2 (weeks 9-12)** adds the guarded generation layer (citations, abstention, audit logging, WORM export) plus graph expansion, as-of retrieval, the bounded multi-step retrieval loop with its per-pass tracing, conversational and profile memory with its ACL re-filtering and multi-turn eval cases, and the nightly evaluation gate, ending in Compliance/Legal/BC sign-off. **Phase 3 (weeks 13-16)** is the full backfill and load testing, bracketed by base backups and closed with the first timed DR drill. **Phase 4** is enterprise rollout, at which point the nightly gates, restore tests, and drills stop being milestones and simply become weather.

v2 candidates wait patiently behind evaluation evidence, in the spirit of the opening bet: LLM-based relationship extraction on high-value collections, thematic clustering with summary pages, and any migration past a scaling gate.

## The Shape of the Thing

Strip away the details, and the whole design is eleven decisions, the first three of which are the principles everything else hangs from:

1. **Simplicity above all.** The fewest moving parts that still deliver the functionality. Simplicity is what makes the project do-able, flexible, maintainable, and affordable: buildable by a small team, prototyped in weeks, run without a fleet of clusters or heavy hardware. In practice that means **no distributed architecture**: one server rather than a cluster, a mounted Unix filesystem rather than an object store, nothing sharded or eventually consistent. Complexity is the project's primary risk, admitted only through measured gates.
2. **Modularity keeps the bets reversible.** Every capability sits behind a contract (retrieval behind an API, sources behind connector plugins, parsers behind the derivative format, models versioned per artifact) so any part can be changed, replaced, or removed without understanding the whole. Loose coupling, clean seams: separate dumplings, never a clump. The same discipline reaches into the code: subdirectories along module boundaries, files under 800 lines, functions under 50, docs at every level, a `README.md` per directory.
3. **AI is a component, not a feature.** Agents run through the whole system (ingestion classification, architecture conformance review, eval clustering, groundedness judging, operations chat, maintainer tooling, the user chat) and through the process that designed it. Each one is built the same way: a knowledge base it must cite, a memory scoped to one identity, and tools reachable only through the documented, audited APIs. The user chat remembers previous requests, so answers get refined instead of retyped, and memory stays context rather than evidence: it never becomes a source of facts, and it never outlives a permission. The agent is available to **every user and every maintainer**, who can talk it into building mini-tools, composing and scheduling workflows, running analyses, and drafting reports and documents inside their own entitlements.
4. **One system of record.** PostgreSQL holds truth (content indexes, entitlements, graph, queue, audit, agent memory) so consistency is a transaction, not a distributed-systems project.
5. **Parse once; everything downstream is disposable.** Originals and derivatives are permanent; chunks, vectors, indexes, and wiki are projections, rebuildable at will. This is what makes every future migration boring, and boring migrations are the good kind.
6. **Search three ways** (meaning, keywords, structure: the graph between documents and the table of contents inside long ones) with time as a dimension, because each method catches what the others miss, and let the agent run the search more than once, bounded at four passes, because the second question depends on what the first one found. The system retrieves small, relevant, authorized context instead of trusting a giant window to sort it out.
7. **Security and provenance are load-bearing.** ACLs filter before ranking; citations verify before display; every answer leaves an immutable trail; saying "I don't know" is a graded skill.
8. **Trust is re-earned continuously.** A pyramid of tests gates every merge; evaluation gates, reconciliation counts, hallucination sampling, and restore drills gate every day. The system proves it still works after every change, or it stops and says so.

9. **Self-healing by design.** Timeouts and retries recover crashed jobs, corrupted files restore from verified snapshots, missed events refetch, degraded indexes rebuild, and every derived artifact can be regenerated from the source of truth. Restarts are ordinary: no durable state sits in process memory, so a rebooted node rejoins, reclaims its expired leases, and carries on. Humans are paged as needed, for the exceptional and never the routine, and nothing touching entitlements or records is ever repaired by guessing.
10. **No failure is orphaned.** Every escalation becomes a monkey: a task with an owner, a status, and a history. People see and manage their own, hand them off or escalate them, and managers see the whole troop. Whatever the machine gives up on, a named human picks up.
11. **Nothing arrives unvetted.** Dependencies are pinned, aged 30 days, and verified, because "latest" is not a version, it's a gamble. Generated code and workflows reach data only through the documented, audited APIs, so ACLs apply automatically, and they pass the same CI, size, and conformance gates as anything a human typed.

A corpus of three million documents, one honest database on one honest server, an agent that remembers what you asked a minute ago without ever letting that memory speak for the record, and a system designed to be *caught* being wrong before a user ever is, and to patch itself up before anyone has to ask. That's the design.

## Appendix A: Requirements Reference

### A.1 Functional

| ID  | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F1  | Ingest documents from multiple enterprise sources (file shares, SharePoint, email archives, document management systems) in PDF, DOCX, PPTX, XLSX, TXT, HTML, Markdown, and common scanned image formats.                                                                                                                                                                                                                                                                                                                                                           |
| F2  | Handle scanned documents via OCR, with extraction confidence scored and attached at the chunk/page level.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| F3  | Preserve document hierarchy: section headers, page and slide numbers, table schemas, footnotes, and speaker notes.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| F4  | Detect and handle exact duplicates, near-duplicates, and superseded versions with alias mapping.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| F5  | Support hybrid retrieval: exact/lexical match (tickers, ISINs, clause numbers, dates, party names) <b>and</b> semantic match (conceptual questions, thematic synthesis, paraphrase), with symmetric pre-processing so queries and chunks are embedded in comparable form: one shared normalizer (Unicode/whitespace, acronym expansion, canonical entities, dates) on both paths, model-specific <code>query: / passage:</code> formatting, heading-path context in chunks, and optional HyDE for broad queries, with each step evaluated against the golden suite. |
| F6  | Filter by rich metadata: source, business unit, document type, date range, classification, and language.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| F7  | Return answers with verifiable citations at the page / slide / section / table level, deterministically linked to the exact document version used.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| F8  | Abstain or flag uncertainty when evidence is weak, contradictory, or superseded. Every answer carries a visible <b>confidence signal</b> derived from retrieval scores, citation coverage, OCR confidence, and (where used) multi-model agreement.                                                                                                                                                                                                                                                                                                                  |
| F9  | Expose retrieval as an independent, secured REST/gRPC service consumable by the agent and other internal systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| F10 | Re-index automatically upon source modification; tombstone deleted content promptly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| F11 | Provide a human-browsable, normalized Markdown corpus for domain expert QA, curation, and compliance audits.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| ID  | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F12 | Maintain a typed inter-document link graph (supersession, amendment, citation, exhibit, shared entities) and an entity index in PostgreSQL, populated deterministically at ingestion and usable for retrieval-time context expansion.                                                                                                                                                                                                                                                                                                                                                         |
| F13 | Render the corpus as an interconnected wiki: Markdown files with YAML frontmatter and <code>[[wikilinks]]</code> (Obsidian-compatible), generated from the database, human-navigable and grep-friendly, scoped to authorized audiences.                                                                                                                                                                                                                                                                                                                                                       |
| F14 | Support point-in-time ("as-of") retrieval via an optional <code>as_of_date</code> parameter, with historical answers explicitly flagged.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| F15 | Provide human interfaces: an operations command center (dashboards and controls for data loads, removals, updates, validation, and testing), an agentic admin chat, and an agentic end-user chat with clickable citations, as-of date selection, and one-click answer flagging into the evaluation queue. All interfaces consume the same audited APIs; state-changing actions require confirmation and are logged.                                                                                                                                                                           |
| F16 | Provide a maintainer task system ("monkeys") as a separate FastAPI web app: per-user task lists with status, history, and failure visibility; reassignment, vacation handoff, and <b>escalation</b> to a group or manager; a manager-wide dashboard over all tasks; read-only visibility for end users into the tasks their own flags created; automatic task creation from dead-letter jobs / eval-gate trips / flagged answers / scrub and reconciliation failures, so no escalation path terminates without an owner.                                                                      |
| F17 | Offer hallucination-reduction answer modes for high-stakes queries: <b>fusion</b> (2 or 3 independent self-hosted models answer from the same retrieved context; a judge model synthesizes, keeping multi-model claims with verified citations and dropping/flagging single-model claims) and <b>persona cross-examination</b> (one model answering and reviewing from distinct perspectives, surfacing divergence). Tiered activation: sensitive collections, low retrieval confidence, or user-requested "verify mode"; effectiveness gated by golden-suite and groundedness evaluation.    |
| F18 | Maintain per-user conversational and profile memory so requests can be refined iteratively. Thread memory (entities, filters, as-of date, documents already shown) rewrites each follow-up into a standalone query, displayed to the user and editable, before it enters the shared normalizer. Profile memory (usual collections, currency preference, answer length, entity disambiguations, previously flagged answers) persists across sessions. Users can view, edit, and delete any memory item, a whole thread, or everything.                                                         |
| F19 | Ground every agent in the system (ingestion classification, conformance review, eval clustering, groundedness judging, admin chat, maintainer tooling, user chat) in the retrieval or admin APIs, with citations where the agent produces user-visible claims, and no direct database access for any agent.                                                                                                                                                                                                                                                                                   |
| F20 | Make the agent's workshop available to <b>all users and maintainers</b> : build mini-tools and saved views, compose and schedule jobs and ETL workflows, run analyses, and generate reports and documents, by asking in plain language. Maintainers additionally get data-plane jobs (backfill, re-parse, re-index); end-user artifacts are read-only per S6. Everything generated reaches data only through the documented, audited APIs, inherits its author's entitlements, and passes the standard CI, size, and AI conformance gates.                                                    |
| F21 | Support <b>bounded multi-step (agentic) retrieval</b> : the agent may issue follow-up searches based on what earlier passes returned, to a maximum of four passes. Every pass applies the full pre-ranking ACL filter independently and is recorded in the query trace with its query, filters, chunk IDs, and scores. Exhausting the budget produces an abstention naming the missing evidence, never a partial answer presented as complete. Activation is conditional and gated by golden-suite evidence.                                                                                  |
| F22 | Support <b>tree-navigation retrieval</b> for long structured documents (PageIndex-style): a per-version section tree (title, page range, summary per node) built from parsed derivatives and stored in PostgreSQL; an LLM descends the tree to locate answering sections within documents shortlisted by hybrid search. Each descent counts as one pass of the bounded loop, inherits document ACLs, and logs its path in the query trace for page-level citation. Activation is conditional, per document type, and gated by golden-suite evidence (including FinanceBench-style questions). |

## A.2 Security, Compliance, and Governance

| ID | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S1 | Access control strictly mirrors source-system entitlements. ACL filters are applied <b>pre-ranking</b> at the database layer; unauthorized users can never retrieve or observe chunks from restricted documents.                                                                                                                                                                                                                              |
| S2 | Information barriers (ethical walls, client/deal segregation) enforced at the tenant / collection partition level.                                                                                                                                                                                                                                                                                                                            |
| S3 | Automated classification tags at ingestion: PII, MNPI, client-confidential, regulatory, and retention category.                                                                                                                                                                                                                                                                                                                               |
| S4 | Full audit log per query: user identity, evaluated roles/groups, question as typed and as rewritten from memory, memory items injected into the prompt, applied filters, retrieved chunk IDs, ranking scores, model/prompt versions, generated response, and rendered citations.                                                                                                                                                              |
| S5 | Complete data residency: originals, derivatives, vector indices, and audit logs remain within the firm's approved security boundary. Models are self-hosted or private enterprise endpoints.                                                                                                                                                                                                                                                  |
| S6 | Release v1 is strictly <b>read-only</b> : no execution of financial transactions, external communications, or automated decisions.                                                                                                                                                                                                                                                                                                            |
| S7 | Prompts, retrieved contexts, and outputs are archived as immutable business records; closed audit-log partitions are exported to write-once storage.                                                                                                                                                                                                                                                                                          |
| S8 | Agent memory is bound to a single user identity, never shared across users, and re-filtered through the pre-ranking ACL join on every read, so revoked entitlements take effect in the next conversational turn. Memory is context only and never a source of asserted facts; it is user-visible, user-erasable, retention-limited (thread 30 days, profile 12 months, both policy-configurable), and disposable without loss of correctness. |

## A.3 Non-Functional

| ID  | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N1  | <b>Corpus Scale:</b> Low single-digit millions of source documents; 20-30 million indexed chunks after deduplication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| N2  | <b>Query Latency:</b> p95 under 1.5 s per retrieval pass (excluding LLM generation); p95 under 4.5 s end-to-end for single-pass answers; p95 under 10 s end-to-end for queries escalated to the bounded multi-step loop, with per-pass progress shown to the user.                                                                                                                                                                                                                                                                                                                                                                                       |
| N3  | <b>Ingestion:</b> Backfill in weeks; incremental sync in minutes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| N4  | <b>Operational Simplicity:</b> No distributed architecture in v1. A single PostgreSQL server (plus a passive standby for failover) and mounted filesystems, with no sharding, no object store, no message broker, and no multi-cluster maintenance overhead.                                                                                                                                                                                                                                                                                                                                                                                             |
| N5  | <b>Reliability:</b> Transactional, observable, crash-resilient job processing with automatic retries under heartbeated leases.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| N11 | <b>Restart Recovery:</b> Any process, node, or the database may restart, planned or otherwise, without human intervention and without data loss beyond in-flight requests. No durable state resides in process memory; interrupted jobs resume from the queue, interrupted conversations resume from stored memory, and interrupted index builds discard cleanly. Full service returns within <b>10 minutes</b> of a full-stack restart, in a scripted order (mounts verified → database crash recovery → workers reclaim leases → ACL sync → model warm-up and readiness → reconciliation), exercised on every deploy and timed in the quarterly drill. |
| N6  | <b>Measurable Quality:</b> Recall@20, citation accuracy, groundedness, and permission-leak rate continuously evaluated against a golden dataset.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| N7  | <b>Portability:</b> Chunk indices are disposable and 100% reconstructible from stored derivatives without re-parsing raw files.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| N8  | <b>Storage Platform:</b> Originals and derivatives on mounted block/file volumes inside the firm's network, with no object-store dependency.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| ID  | Requirement                                                                                                                                                                                                                                                           |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N9  | <b>Backup &amp; Recovery:</b> RPO $\leq$ 1 hour, RTO $\leq$ 8 hours; immutable backups inside the security boundary; automated restore tests at least monthly.                                                                                                        |
| N10 | <b>Lifecycle &amp; Continuous Evaluation:</b> Content changes visible within minutes; ACL revocations within 5 minutes; nightly evaluation gate verifies quality, completeness, and hallucination level after each day's updates and freezes ingestion on regression. |

## Appendix B: Technology Choices

| Layer                   | Selected Technology                                                                                                                                                                                                                                                 | Why                                                                                                                                                                                                                                                            |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Originals & derivatives | Mounted volumes (EBS/NFS or NAS/SAN), XFS + LVM, content-addressed and write-once                                                                                                                                                                                   | In-boundary, snapshot-consistent by construction, POSIX-fast for parsers                                                                                                                                                                                       |
| Parsing                 | Docling (heavy path) + PyMuPDF (fast path)                                                                                                                                                                                                                          | Layout/table fidelity with OCR confidence; 10× faster path for born-digital files                                                                                                                                                                              |
| Database & indexing     | PostgreSQL 16+ with <code>pgvector</code> (HNSW, <code>halfvec</code> ) and <code>pg_search</code> (BM25)                                                                                                                                                           | One engine for ACL joins, keyword search, vector search, queue, and audit                                                                                                                                                                                      |
| Document graph & wiki   | Plain PostgreSQL edge/entity tables; wiki rendered as Markdown + frontmatter + wikilinks (Obsidian-compatible)                                                                                                                                                      | Structure stays in the system of record; wiki is a disposable projection                                                                                                                                                                                       |
| Job queue               | PostgreSQL <code>SKIP LOCKED</code>                                                                                                                                                                                                                                 | Transactional, broker-free                                                                                                                                                                                                                                     |
| Agent memory            | Plain PostgreSQL tables (thread memory and per-user profile memory), keyed by user identity, ACL-joined on read, retention-bounded, exposed through the retrieval API and a memory panel in the chat UI                                                             | Same system of record as everything else, so a permission change and a memory read can never disagree; disposable projection, audited like any other prompt input                                                                                              |
| Embeddings              | Self-hosted open models (BGE / E5 / Nomic class) on local GPUs; shared query/chunk normalizer module (versioned like the model); asymmetric <code>query:</code> / <code>passage:</code> prefixes; optional HyDE behind a golden-suite gate                          | Data boundary; no per-token fees across 10B+ tokens; queries and chunks embedded in comparable form via one code path that cannot drift                                                                                                                        |
| Document trees          | Plain PostgreSQL <code>doc_tree_nodes</code> table built from Docling heading hierarchy, LLM node summaries once per version; LLM tree descent as an evaluated escalation. PageIndex (VectifyAI, open source) as reference implementation, not a default dependency | Reasoning over a document's own structure finds sections that similarity misses in long filings and agreements; no new engine, just one more disposable derivative                                                                                             |
| Reranker                | BGE-Reranker-Large cross-encoder; LLM rerank only as evaluated escalation                                                                                                                                                                                           | Precision on multi-hop clauses within the latency budget                                                                                                                                                                                                       |
| Retrieval API           | Python FastAPI / <code>asyncpg</code> ; single-pass by default, bounded multi-step (max 4 passes) on escalation, each pass independently ACL-filtered and traced                                                                                                    | Lightweight, typed, decoupled from agents. Python over Rust: the heavy lifting runs in C/GPU libraries (Postgres, CUDA, PyMuPDF), the parsing/ML ecosystem is Python-first, and any measured CPU-bound module can later be swapped to Rust behind its contract |

| Layer                                               | Selected Technology                                                                                                                                                                                                                                                                                                                                                                           | Why                                                                                                                                                                                                      |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend<br>(command center, admin chat, user chat) | Vanilla JavaScript ES modules, no frameworks, no build step; all styles in a single <code>styles.css</code> ; static files served from existing infrastructure; consumes only the documented, audited APIs                                                                                                                                                                                    | Simplicity: no framework dependency to quarantine or migrate; modular per the code rules (one module per feature, README per directory); ACLs and audit apply identically to humans and agents           |
| Maintainer workbench<br>("monkeys")                 | Separate FastAPI app + vanilla-JS frontend; monkey tables in PostgreSQL; auto-created from dead-letter/eval/flag/scrub events; per-user lists with handoff and escalation, manager-wide dashboard, read-only view for the user who raised the flag; agent-built mini-tools and scheduled ETL workflows running through the shared job queue, gated by the standard CI + AI conformance review | Retrieval service stays lean while the workbench evolves at business speed; every escalation path gets an owner, a status, and a history, so no failure is orphaned                                      |
| Toolchain & dependencies                            | Homebrew-installed Python (pinned major.minor); <code>uv</code> with <code>exclude-newer = "30 days"</code> and <code>prerelease = "disallow"</code> ; PGDG-pinned PostgreSQL; Docker images by digest via scanned internal registry                                                                                                                                                          | Supply-chain safety: aged, pinned, verified dependencies; no <code>:latest</code> , no day-old packages                                                                                                  |
| Testing                                             | pytest (unit + module tests via public APIs with faked ports); containerized PostgreSQL for integration tests; fixture corpus for pipeline tests; <code>import-linter</code> + size/docstring checks in CI; AI-agent architecture-conformance review gating merges                                                                                                                            | Every layer guarded: functions, module contracts, real adapters, answer quality (golden suite), and the architecture rules themselves                                                                    |
| LLM inference                                       | Private enterprise endpoint / self-hosted vLLM; optional fusion mode (2 or 3 independent models + judge synthesis) and persona cross-examination for high-stakes queries, tiered by collection sensitivity, retrieval confidence, and user request                                                                                                                                            | Inside the VPC perimeter; independent models rarely agree on the same hallucination, so multi-model consensus filters idiosyncratic errors at a known 3-4× cost, paid only where the verdict is in doubt |
| Backup & recovery                                   | pgBackRest PITR + standby; locked cross-AZ snapshots; WORM audit exports                                                                                                                                                                                                                                                                                                                      | One-hour RPO; ransomware-resistant; restore path proven by drills                                                                                                                                        |

## Appendix C: Risk Register (Condensed)

| Risk                                               | Mitigation                                                                                                                                                                                                               |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Permission leakage (incl. via graph links or wiki) | Pre-ranking SQL ACL joins everywhere including graph hops; wiki rendered only in curator enclave; adversarial leak tests in CI; 0.00% tolerance                                                                          |
| Permission leak through conversation memory        | Memory bound to one identity, re-filtered by the pre-ranking ACL join on every read, never shared between users or sessions; adversarial multi-turn inheritance cases in the golden suite; memory items logged per query |
| Stale or misleading memory steering answers        | Memory is context, not evidence: every claim cites a chunk retrieved this turn; the rewritten question is shown and editable; retention windows expire thread and profile memory; users can delete any item              |
| Missing specific clauses                           | Adaptive routing lets identifier queries bypass summary filtering; the multi-step loop can go back for what a single pass missed                                                                                         |

| Risk                                                                                    | Mitigation                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multi-step retrieval burning latency or compounding its own errors                      | Hard cap of four passes; per-pass ACL filter and trace; budget exhaustion produces an abstention naming the gap, not a partial answer; escalation enabled only where golden-suite evidence justifies the latency                                                                                                |
| Stale or wrong-period answers                                                           | Validity windows, as-of filtering, temporal/currency ranking boosts, supersession flags, as-of golden cases                                                                                                                                                                                                     |
| Hallucinated citations / fabricated claims                                              | Middleware validates every citation token against retrieved chunks; unanswerable-set testing; nightly judge sampling; fusion mode (multi-model consensus + judge synthesis) and persona cross-examination for sensitive or low-confidence queries                                                               |
| Silent ingestion failure / stale index                                                  | Nightly three-layer reconciliation; index-lag metric; self-retrieval probes                                                                                                                                                                                                                                     |
| Restart or reboot losing in-flight work, or a node returning in an inconsistent state   | No durable state in process memory; heartbeated leases reclaimed on start; idempotent steps; shadow-table index builds; scripted boot order with ACL sync before first query and readiness gated on a real answer; post-restart reconciliation; rolling restarts proven on every deploy and timed quarterly     |
| Routine transient failures (crashed workers, corrupt files, missed events, index bloat) | Self-healing: idempotent retries with backoff, automatic quarantine-and-restore from verified snapshots, reconciliation-driven refetch, automatic partition reindex, regeneration of derived artifacts; dead-letter + page only after retries are exhausted; entitlements and records always escalate to humans |
| Quality regression after updates                                                        | Nightly golden-suite gate vs. 7-day baseline; page + ingestion freeze on trip                                                                                                                                                                                                                                   |
| Architecture erosion in code                                                            | <code>import-linter</code> + size/docstring checks in CI; AI-agent conformance review on every merge                                                                                                                                                                                                            |
| Agent-generated tools or workflows bypassing controls (whoever asked for them)          | Generated artifacts reach data only through documented APIs, so ACLs and audit apply automatically and a generated page cannot see what its author can't; end-user artifacts are read-only per S6; everything passes the same CI, size, and conformance gates as human-written code                             |
| Document volume loss/ corruption                                                        | Locked cross-AZ snapshots; write-once discipline; hash scrub                                                                                                                                                                                                                                                    |
| PostgreSQL loss / logical corruption                                                    | Continuous WAL archiving + PITR; standby for failover; monthly restore tests                                                                                                                                                                                                                                    |
| Audit record loss                                                                       | Monthly partitions exported nightly to WORM with hash manifests; records-policy-only deletion                                                                                                                                                                                                                   |
| Backup destruction (ransomware / rogue admin)                                           | Snapshot lock, separate backup account, WORM archive, scrub-verified clean-point restore                                                                                                                                                                                                                        |
| Untested restore path                                                                   | Monthly automated restore tests page on failure; quarterly timed DR drills reported to Compliance/BC                                                                                                                                                                                                            |